

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 4: Quadratic Equations

Previous year board questions, topic-wise: 2 questions

The tag after each question gives the board paper and the marks. Solutions are in a separate PDF with the same question numbers.

1 Nature of roots

1. If the roots of the equation $ax^2 + bx + c = 0$, $a \neq 0$, are real and equal, then which of the following relations is true? (a) $a = \frac{b^2}{c}$ (b) $b^2 = ac$ (c) $ac = \frac{b^2}{4}$ (d) $c = \frac{b^2}{a}$ **CBSE 2024, 1 mark**

2 Word problems

2. A rectangular floor area can be completely tiled with 200 square tiles. If the side length of each tile is increased by 1 unit, it would take only 128 tiles to cover the floor. (i) Assuming the original length of each side of a tile to be x units, make a quadratic equation from the above information. (ii) Write the corresponding quadratic equation in standard form. (iii) Find the value of x by factorisation, OR solve the equation using the quadratic formula.

CBSE 2024, 4 marks, case study